

THE WORLD IN TWENTY TRIANGLES

DYMAXION PROJECTION

A DYMAXION MAP TO CUT, FOLD AND GLUE

PRINT MORE COPIES

Scan for letter and 11 x 17 sheets
to print, cut and fold again.



FLAT EARTH / BUTTERFLY MAPS



DIRECTION



DISTANCE

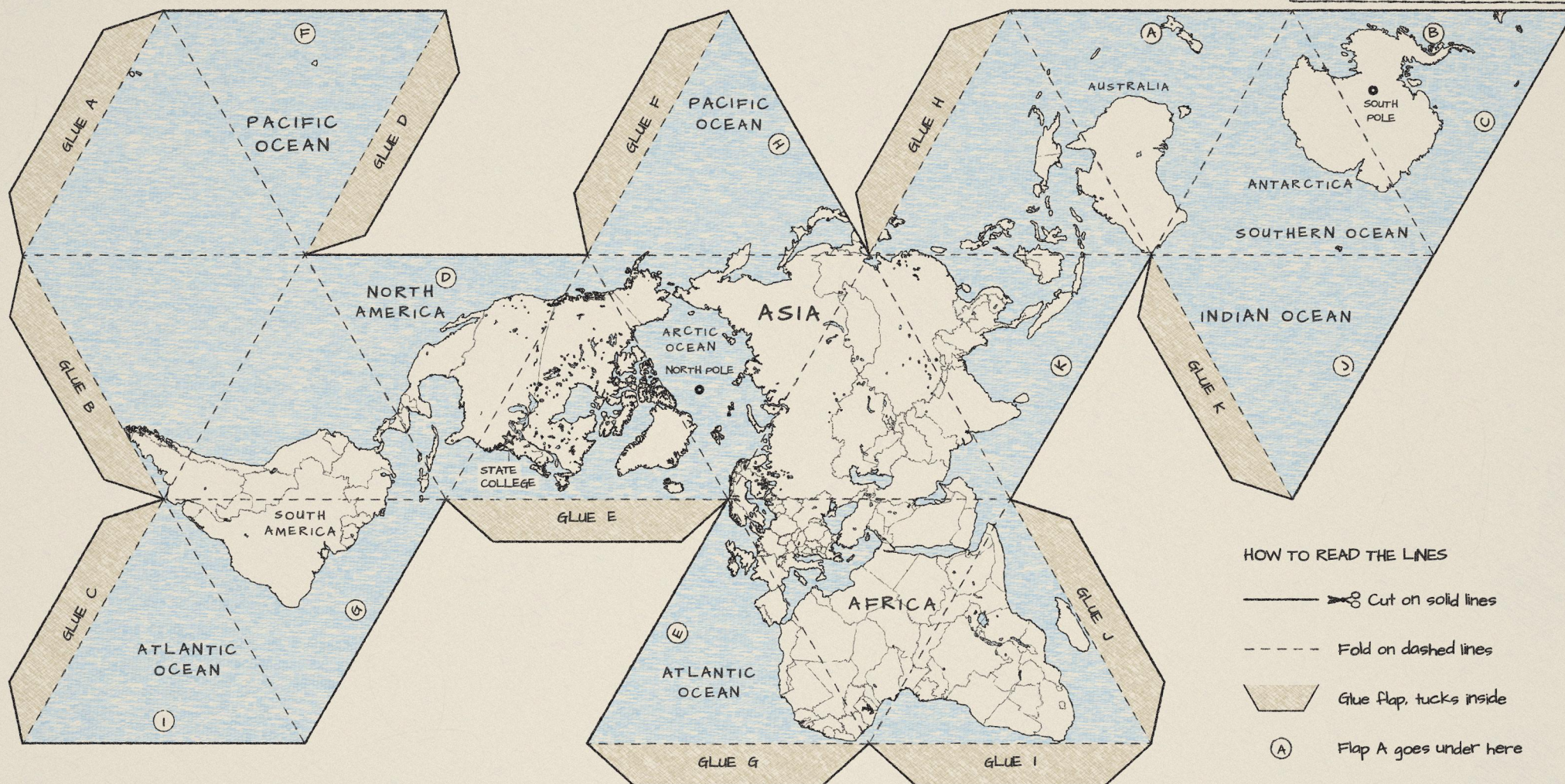


SIZE



SHAPE

A Dymaxion map (Fuller, 1954).
Sizes stay within about 10 percent.
Shapes bend a little near the corners.
Green: true everywhere.
Half green: true only in places. Red: lost.



HOW TO READ THE LINES

— Cut on solid lines

- - - Fold on dashed lines

Glue flap, tucks inside

(A) Flap A goes under here

FULLER'S MAP

The inventor Buckminster Fuller first drew his map in 1943 on a shape with 14 faces. In 1954 he moved it onto an icosahedron: twenty equal triangles that fit together into a ball. No butterfly map in this book has more faces, so this one is the roundest.

Each triangle covers only a small piece of the Earth, so flattening it hardly stretches anything. More faces, less stretch: sizes stay within about 10 percent of true, and shapes bend only a little near the corners.

Like Cahill before him, Fuller put every cut in the ocean, so the land stays in one piece. He called it one island in one ocean. His map has no top or bottom: turn the page any way you like and it is still right.

The price is the gaps. The oceans are split into pieces, and a straight line only stays straight inside one triangle.

Dymaxion is a made-up word, from dynamic, maximum and tension. Fuller gave the same name to a house, a car and a bathroom he designed.

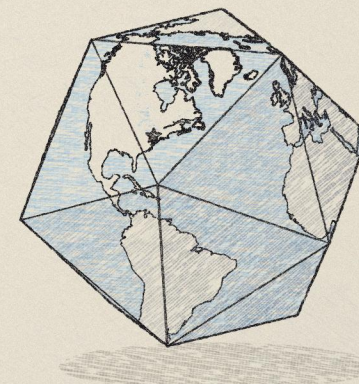
HOW TO MAKE IT

- 1 COPY IT
Scan the code at the top to print fresh copies, or photocopy this page. Thin cardstock works best. That way your book stays whole.
- 2 CUT
Cut around the outside on the solid lines, flaps and all. Never cut a dashed line.
- 3 FOLD
Crease every dashed line, folding the paper away from you so the map stays on the outside.
- 4 GLUE
Put glue on a flap and press it under the edge with the same letter. A glue stick works best.
- 5 CLOSE IT
Save the last flaps for the end. Glue them, then press gently from outside with a finger.

YOU WILL NEED

- Scissors and a glue stick
- A ruler and an empty ballpoint pen, to press a groove along each dashed line so it folds crisply

WHAT YOU WILL MAKE



A 20-sided globe, folded from the net.

TRY THIS!

Turn your globe so the star at State College is on top. What is on the bottom? Then find the North Pole. Which way is up now?